

M378K: September 2<sup>nd</sup>, 2026.

M378K Introduction to Mathematical Statistics

Problem Set #3

Named discrete random variables.

**Problem 3.1.** Source: Sample P exam, Problem #125.

An electronic system contains three cooling components that operate independently. The probability of each component's failure is 0.05. The system will overheat if and only if at least two components fail. Calculate the probability that the system will overheat.

→:  $Y$ ... # of components that fail

$P[Y \geq 2] = ?$

$$Y \sim b(n=3, p=0.05)$$

$P[Y \geq 2] \stackrel{\text{Idea 1.}}{=} P_Y(2) + P_Y(3)$

$$= \binom{3}{2} (0.05)^2 (0.95)^1 + (0.05)^3$$

$$= (0.05)^2 (3 \cdot 0.95 + 0.05) = 2.9 (0.05)^2$$

2.85

$\stackrel{\text{Idea 2.}}{\rightarrow} 1 - P_Y(0) - P_Y(1) = \dots = \boxed{0.00725}$



## Geometric Dist'n.

We repeat **independently** Bernoulli trials w/ the same success probability  $p$  until the first success.

The random variable  $Y$  which denotes the number of failures until the first success is called the **geometric distribution w/ parameter  $p$ .**

$$Y \sim g(p)$$

Set  $q := 1 - p$

| $y$      | 0   | 1           | 2             | ... | $k$           | ... |
|----------|-----|-------------|---------------|-----|---------------|-----|
| $p_Y(y)$ | $p$ | $q \cdot p$ | $q^2 \cdot p$ | ... | $q^k \cdot p$ | ... |

Q:  $\mathbb{P}[Y > 2] = 1 - \mathbb{P}[Y \leq 2]$

$$= 1 - p_Y(0) - p_Y(1) - p_Y(2)$$

$$= 1 - p - qp - q^2 p$$

Idea 1.  $= 1 - p(1 + q + q^2)$

$$= 1 - p \cdot \frac{1 - q^3}{1 - q}$$

$$= 1 - (1 - q^3) = q^3$$

Idea 2.  $= 1 - p - qp - q^2 p$

$$= q - qp - q^2 p$$

$$= q(1 - p - qp)$$

$$= q(q - qp)$$

$$= q \cdot q(1 - p) = q^3 \quad \square$$

In general

$$\mathbb{P}[Y > m] = q^{m+1}$$



**Problem 3.2.** Source: Sample P exam, Problem #462.

Each person in a large population independently has probability  $p$  of testing positive for diabetes where  $0 < p < 1$ . People are tested for diabetes, one person at a time, until a test is positive. Individual tests are independent. Determine the probability that  $m$  or fewer people are tested, given that  $n$  or fewer people are tested, where  $1 \leq m \leq n$ .

→:  $Y'$  ... total number of people tested  
       ↳ SHIFTED geometric w/ parameter  $p$ ,  
       i.e.,  $Y = Y' - 1 \sim g(p)$

$$\begin{aligned} \mathbb{P}[Y' \leq m \mid Y' \leq n] &= \mathbb{P}[Y+1 \leq m \mid Y+1 \leq n] \\ &= \frac{\mathbb{P}[Y+1 \leq m, Y+1 \leq n]}{\mathbb{P}[Y+1 \leq n]} \\ &= \frac{\mathbb{P}[Y \leq m-1, Y \leq n-1]}{\mathbb{P}[Y \leq n-1]} \\ &= \frac{\mathbb{P}[Y \leq m-1]}{\mathbb{P}[Y \leq n-1]} \\ &= \frac{1 - \mathbb{P}[Y > m-1]}{1 - \mathbb{P}[Y > n-1]} \\ &= \frac{1 - q^m}{1 - q^n} \quad (\star) \end{aligned}$$

## The Poisson Dist'n.

We say that  $Y$  is Poisson (dist'd) w/ parameter  $\lambda$   
 and write  $Y \sim \text{Poisson}(\lambda) \sim \mathcal{P}(\lambda)$   
 if its pmf is

$$p_k = p_Y(k) = e^{-\lambda} \cdot \frac{\lambda^k}{k!} \quad \text{for all } k=0, 1, \dots \quad (\text{i.e., } k \in \mathbb{N}_0)$$